

Music Performance & Streaming Analytics

1. Project Overview and Objective

The Music Performance and Streaming Analytics project focuses on analyzing music consumption data from multiple streaming platforms to understand song performance, audience engagement, and emerging trends. The project collects and evaluates key metrics such as streams, playlist placements, listener reach, Shazam counts, station plays, and platform-specific performance indicators from services such as Spotify, Apple Music, Amazon Music, Deezer, Pandora, and SoundCloud.

By applying data analysis and visualization techniques, the project provides insights into artist growth, track popularity, listener behaviour, and the effectiveness of music promotion strategies. The analytics dashboard helps stakeholders monitor performance trends, compare platforms, and make data-driven decisions for marketing, distribution, and audience development.

2. Data Sources

- **Source Description and Timeline:** <https://www.kaggle.com/datasets>
- **Domain:** Music Performance

3. Problem Statement

This project aims to analyze a music streaming dataset to identify the key factors influencing song popularity, compare performance across streaming platforms, evaluate artist and album success, and uncover trends based on release year and user engagement. The analysis will help answer business questions related to streaming performance, audience behavior, and the characteristics of successful songs through data cleaning, exploratory analysis, and interactive visualizations.

- Lack of real-time insights into music performance quality
- Artist wise spotify playlist count
- Number of spotify playlist count
- Number of time viewers clicked the like button
- Number of time tiktok videos have been watched
- Number of apple music playlist that contain a particular song

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
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Track	String	A track usually means one individual song
Album Name	String (Text)	The title of that collection
Artist	String (Text)	Name of the artist
Release Date	Date	Date of releasing
ISRC	String (Text)	Tracks music plays
All time Rank	Number	the position of a song, album, or artist compared with all others throughout history
Track score	Number	numerical value that measures how well a song (track) is performing.
Spotify Streams	String (Text)	each time a song is played for at least 30 seconds
Spotify Playlist Count	Number	number of Spotify playlists that include a particular song
Spotify Playlist Reach	Number	total potential audience of all the playlists that include a song
Spotify Popularity	Number	Spotify assigns to a track, album, or artist to indicate how popular it is at the current time
Youtube Views	Number	show how many times a song's official video, lyric video, or audio video has been watched
Youtube likes	Number	number of times viewers have clicked the Like button on a YouTube video
TikTok post	Number	video or piece of content uploaded to TikTok by a user, creator, or artist
TikTok likes	Number	number of times users have tapped the Like button on TikTok videos
TikTok views	Number	number of times TikTok videos have been watched

Youtube Playlist Reach	Number	potential number of viewers who can discover or watch a song through YouTube playlists that include it
Apple music playlist count	Number	number of Apple Music playlists that contain a particular song.
Air play spins	Number	AirPlay keeps showing a loading/spinning circle and does not finish connecting.
SiriusXM spins	Number	SiriusXM keeps showing a loading/spinning icon and does not start playing or finish connecting.
Deezer playlist count	Number	number of playlists associated with an account, artist, or search result.
Deezer playlist reach	Number	number of people who can potentially be reached through a placement on Deezer.
Amazon playlist count	Number	number of playlists on Amazon Music, there is no public page that shows a total “Amazon playlist count.”
Pandora streams	Number	number of times a song, album, or artist has been played on Pandora.
Pandora track stations	Number	number of Pandora stations generated from a particular track
Sound cloud streams	Number	number of times a track has been played on SoundCloud.
Shazam counts	Number	number of times a song has been identified using Shazam.

5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.

- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

- **Data Cleaning & Transformation:** Removed duplicates, handled missing values, standardized formats, and created measures.

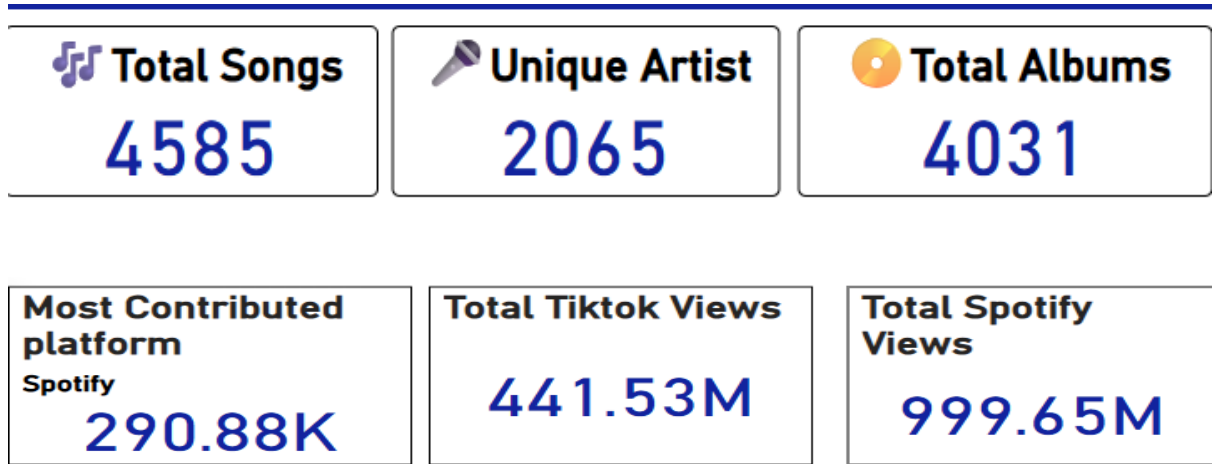
Track	Album Name	Artist	Release Date
A Bar Song (Tipsy)	A Bar Song (Tipsy)	Shaboozey	04-12-2024
Espresso	Espresso	Sabrina Carpenter	04-12-2024
Million Dollar Baby -Single	Million Dollar Baby -Single	Tommy Richman	26-04-2024
Not Like Us	Not Like Us	Kendrick Lamar	05-04-2024
I Like The Way You Kiss Me	I Like The Way You Kiss Me	Artemas	19-03-2024
Gata Only	Gata Only	Floyymenor	02-02-2024
Lunch	Hit Me Hard And Soft	Billie Eilish	17-05-2024
Birds Of A Feather	Hit Me Hard And Soft	Billie Eilish	17-05-2024
I Had Some Help (Feat. Morgan Wallen)	I Had Some Help	Post Malone	05-10-2024
Chihiro	Hit Me Hard And Soft	Billie Eilish	17-05-2024
Tell Ur Girlfriend	Tell Ur Girlfriend	Lay Bankz	02-07-2024
End Of Beginning	End Of Beginning	Djo	16-09-2022
Fortnight (Feat. Post Malone)	The Tortured Poets Department	Taylor Swift	18-04-2024
Houdini	Houdini	Eminem	31-05-2024
Like That	We Don'T Trust You	Future	22-03-2024
Cruel Summer	Cruel Summer	Taylor Swift	23-08-2019
Greedy	Greedy	Tate Mcrae	14-09-2023
As It Was	As It Was	Harry Styles	31-03-2022
We Can'T Be Friends (Wait For Your Love)	Eternal Sunshine	Ariana Grande	03-08-2024
Lose Control	Lose Control	Teddy Swims	23-06-2023
Saturn	Saturn - Single	Sza	24-02-2024
Stargazing	Stargazing	Myles Smith	05-10-2024
Whatever She Wants	Whatever She Wants	Bryson Tiller	13-02-2024
Austin	Austin	Dasha	17-11-2023
Wildflower	Hit Me Hard And Soft	Billie Eilish	17-05-2024
Good Luck, Babe!	Good Luck, Babe!	Chappell Roan	04-05-2024

- **Pivot Tables:** Generated Pivot Tables for data summarisation and initial insights.

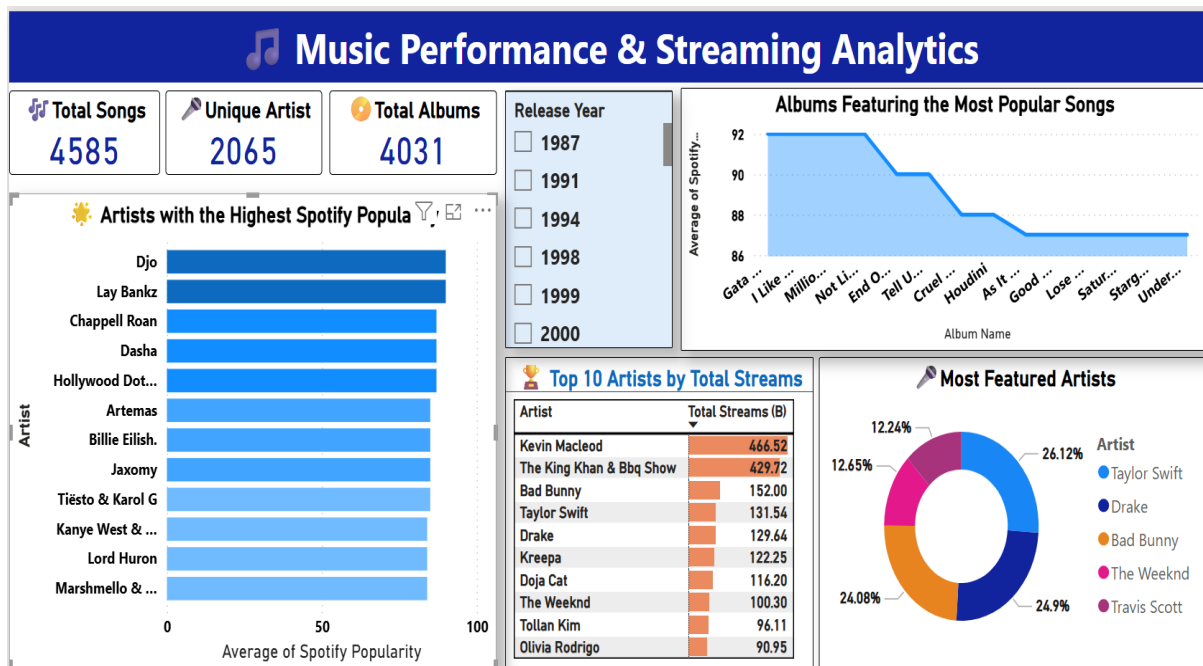
Top 10 Artist Performance		Most frequent artist in the dataset	
Artist	Sum of Total View	Artist	Count of Artist
Bad Bunny	152.00B	Taylor Swift	64
Doja Cat	116.20B	Drake	61
Drake	129.64B	Bad Bunny	59
Kevin Macleod	466.52B	The Weeknd	31
Kreepa	122.25B	Travis Scott	30
Olivia Rodrigo	90.95B	Karol G	28
Taylor Swift	131.54B	Ariana Grande	26
The King Khan & Bbq Show	429.72B	Billie Eilish	26
The Weeknd	100.30B	Future	23
Tollan Kim	96.11B	Post Malone	21
		Juice Wrld	21
		Peso Pluma	21
		Olivia Rodrigo	21
		Morgan Wallen	20
		Lil Baby	20
Highest Popularity			
Artist	Average of Spotify Popularity		
Lay Bankz	90		
Djo	90		
Chappell Roan	87		
Dasha	87		
Hollywood Dots The Oncelor	87		
Tiësto & Karol G	85		
Jaxomy	85		
Billie Eilish.	85		
Artemas	85		
Lord Huron	84		
Kanye West & Ty Dolla Sign	84		
Marshmello & Fuerza Regida	84		

7. Data Modelling and DAX (Power BI)

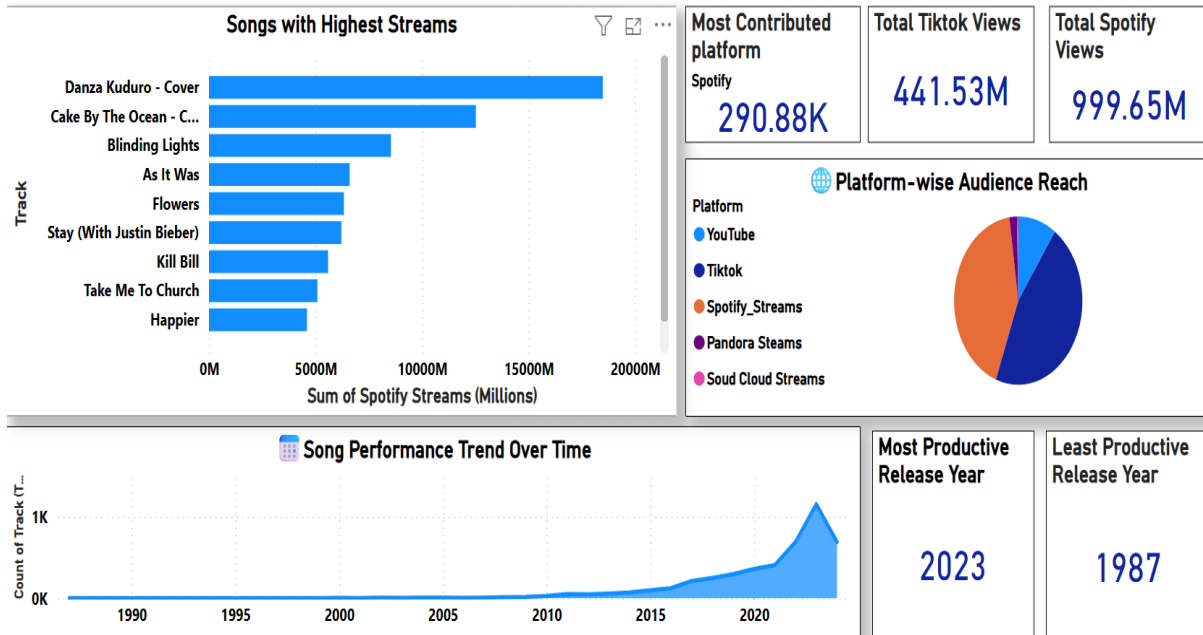
- **Calculated Columns & DAX Measures:** Implemented DAX formulas for key metrics, such as total songs, unique artists and total no. of albums



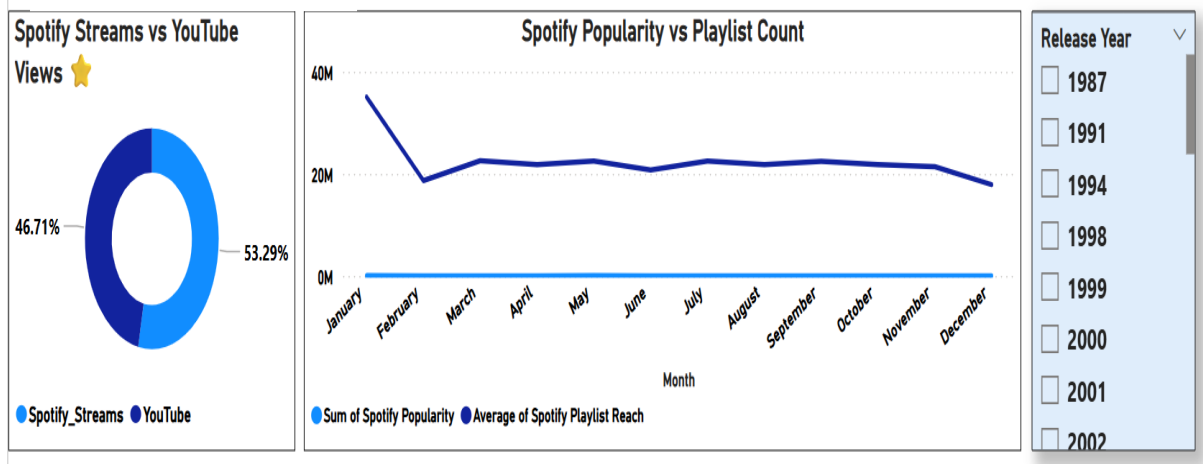
8. Analysis and Visualizations (Power BI)



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9. Insights & Conclusions

- The dataset contains 4,585 songs, 2,065 unique artists, and 4,031 albums, providing a comprehensive view of music streaming performance.
- A small group of artists accounts for a significant share of total streams and popularity, highlighting the concentration of audience engagement.
- Spotify and TikTok are the leading platforms in terms of audience reach, indicating their importance for music promotion.
- The number of song releases has increased steadily over the years, with **2023** emerging as the most productive release year.
- Albums featuring highly popular songs generally achieve better overall performance.

- Cross-platform analysis suggests a positive relationship between Spotify streams and YouTube views, indicating consistent audience engagement across platforms.
 - Playlist inclusion appears to improve song visibility and is associated with higher Spotify popularity.
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10. Conclusions

Descriptive Analysis (What happened?)

The dataset contains 4,585 songs, 2,065 artists, and 4,031 albums, with a few artists and songs contributing the majority of streams.

The descriptive analysis provides a comprehensive overview of the music streaming dataset by summarizing key performance indicators, identifying top-performing artists, songs, and albums, comparing audience reach across platforms, and highlighting release trends over time. These findings establish a strong foundation for deeper diagnostic, predictive, and prescriptive analyses.

Diagnostic Analysis (Why did it happen?)

Spotify streams and YouTube views show a positive relationship.

Playlist inclusion helps improve song popularity.

Popular songs increase an album's overall performance.

Multi-platform presence increases audience reach and engagement

The number of song releases has increased steadily over the years, with 2023 emerging as the most productive release year but in 2024, songs release is slightly decreased.

Predictive Analysis

Historical trends suggest that popular artists, playlist inclusion, and promotion across platforms like Spotify, YouTube, and TikTok can help songs achieve better streaming performance and reach a wider audience in the future.

Prescriptive Analysis (What Should Be Done?)

Prescriptive analysis recommends actions based on the insights obtained from the descriptive, diagnostic, and predictive analyses. It helps identify strategies that can improve song performance, increase audience reach, and support better marketing decisions.

- 1) Promote songs across **Spotify, YouTube, TikTok, and other streaming platforms** to maximize audience reach and engagement.
- 2) Collaborating with well-known artists can help increase song visibility, attract new listeners, and improve streaming performance.
- 3) Since **Spotify and TikTok** contribute the largest audience reach, marketing efforts should prioritize these platforms to maximize impact.

- 4) Promote New Releases
- 5) Use streaming data to optimize future music marketing strategies